

Table 1. Comparison of three commonly used multiattribute utility scales

MAUS	Health dimensions	Levels/ dimension	Total number of health states
QWB	3 functioning dimensions: mobility, physical activity, social activity + 27 symptoms/problems	3 2	1170
HUI-III	8 dimensions: vision, hearing, speech, ambulation, dexterity, emotion, cognition, pain/discomfort	5–6	972 000
EuroQol	5 dimensions: mobility, self-care, usual activity, pain/discomfort, anxiety/depression	3	243

Source: Adapted from [16].

(e.g. hospitalization and unable to perform normal activities). A crucial assumption in developing the composite (total) QWB score is that it is an additive combination of the four subcomponent scores. If each of these subcomponents contributes equally to utility, then, as measures of changes in severity, they should also have the same impact on WTP. Through a meta-analysis of WTP estimates, it is possible to test this restriction.

QALYs take health measurement one step further, by combining information on severity (typically from MAUS or related utility scores) and duration in a very specific way. According to the QALY framework, the health-related quality of life corresponding to any possible health state ($i = 1$ to N) can be represented by a single utility index or weight (q_i), generally varying between 0 (death) and 1 (perfect health), although in some instances weights less than 0 (worse than death) are also considered. These 'QALY weights' are typically derived from SG, TTO, or VAS scoring approaches or from MAUS methods that use these techniques. If T_i presents the number (or fraction) of life years spent in each health state, i , then the number of QALYs corresponding to a lifetime health profile can be expressed as

$$\text{QALY} = \sum_{i=1}^N q_i T_i \quad (5)$$

When QALYs are used as a representation of individual preferences, they imply that the utility (or disutility) associated with time spent in any health state is directly proportional to the duration of the health state. That QALYs are defined to be independent of income (and thus consumption) is in some instances viewed as an advantage because they treat health benefits equitably across any given population (i.e. without regard to individual

economic circumstances). However, independence from income also imposes strong restrictions for interpreting QALYs as a representation of preferences [17].

The assumptions underlying the QALY model also define hypotheses regarding how changes in severity and duration should affect values. Under the most restrictive set of assumptions where QALYs are interpreted as a utility function and consumption is constant over time [18], WTP should increase in direct proportion to the gain in QALYs. This implies the following expression:

$$\text{WTP} = \alpha(\Delta\text{QALY})^\beta \quad \text{where } \beta = 1 \quad (6)$$

In this expression, α can be interpreted as a constant value per QALY gain. In practice, a constant value per QALY or 'QALY valuation' approach has been frequently used (see, for example, [4,5]) for valuing national-level health changes. In most of these applications, the value per QALY is derived by annualizing empirical estimates of the value per statistical life (VSL) [18], resulting in values of roughly \$100 000 per QALY.

Maintaining the constant value per QALY assumption, if the QALY change is composed of a specific change in health-related quality of life (Δq) over a specific time period (Δt), then Equation (6) can be further decomposed into the following expression:

$$\text{WTP} = \alpha(\Delta q \Delta t)^\beta \quad \text{where } \beta = 1 \quad (7)$$

This formulation implies that the elasticities of WTP with respect to Δq and Δt are both equal to 1. Through a meta-analysis of WTP estimates, it is possible to test this hypothesis directly.

As shown by Hammitt [How much is a QALY worth? Utility functions for health and wealth. Unpublished manuscript, 2002.] and Klose [19], under somewhat less restrictive conditions, it is

possible to include QALYs (or QALY weights) in a utility theoretic framework without imposing assumptions that imply a constant WTP per QALY. In other words, QALY weights that are invariant to wealth and longevity can be derived from plausible preference structures that include both health and consumption. Under these conditions, WTP is systematically but not linearly related to QALY gains. In particular, under most plausible conditions, WTP per QALY gain is declining with respect to health status and the size of the QALY gain.

Study population characteristics

Values for health changes are also expected to depend importantly on the characteristics of the individual in question, in particular income, prices, age, and health status.

Income and wealth are primarily expected to have a positive effect on health values. First, individuals' WTP for health changes is constrained by their available budget; therefore, other things equal and as long as health is a normal good, it is expected that individuals with higher incomes will also have higher values for health changes. Second, to the extent that higher incomes imply higher opportunity costs of illness (e.g. through lost wages or household production), they create a price effect. A higher price of illness is expected to increase WTP for health. Higher prices of mitigating goods, such as medication and treatment, also affect the direct cost of illness and are expected to increase the WTP to avoid or reduce morbidity. It is important, however, to emphasize that these prices may be partially or totally covered by employers or health insurers, in which case the marginal effect of these prices on private WTP should be smaller.

An individual's age can also potentially affect his or her WTP for health improvements in a number of ways, but for avoided acute effects it is generally expected to have a positive effect. First, in general terms, age is negatively related to health status. To the extent that marginal utility for improved health status is declining ($\partial^2 V / \partial H^2 < 0$), and age serves as an indicator (inverse) of health status, a positive relationship between WTP and age is expected. Second, age may capture aspects of socio-economic status that are not captured by annual income measures. For example, elderly individuals may have higher accumulated wealth

relative to younger individuals with comparable income. If wealth measures are not adequately controlled for, then age may increase WTP for health improvements through a wealth effect.

Valuation method

The magnitude of value estimates for changes in health is also expected to depend on the way in which they are estimated. All the WTP estimates included in our analysis are based on stated preference (SP) methods (either the contingent valuation method (CVM) or conjoint analysis). Even so, there are a number of methodological differences that have the potential to influence WTP.

One potentially significant difference is the type of value elicitation format used – open-ended (OE) or dichotomous choice (DC). The relative advantages of these formats have been widely discussed and analyzed. Carson [20], for example, argues in favor of the DC format. He suggests that individuals will tend to understate their WTP with the OE format because respondents do not have the incentive and are not accustomed to conditions where they have to 'find' their maximum point. However, if respondents are answering questions strategically, OE responses may overstate or understate WTP [21]. Others have argued that the DC format is susceptible to 'yea-saying,' in which case respondents exhibit an inherent tendency to agree with questions [22–25]. Therefore, the expected direction of this effect is indeterminate, but meta-analysis offers an approach for testing whether systematic differences exist between DC- and OE-based estimates of health values.

Another potentially important difference has to do with how the health change (i.e. the 'commodity') is described. As previously discussed, the valuation context can be framed as either WTP to improve health (CV) or WTP to avoid a decline in health (EV). Even for equivalent sized gains or losses in health (i.e. holding ΔH constant), they will not necessarily have the same value. In particular, if the marginal utility of health decreases with H ($\partial^2 V / \partial H^2 < 0$) or if the marginal utility of income increases with H ($\partial^2 V / \partial Y \partial H > 0$), EV is generally expected to be greater than CV.

Nonetheless, for small changes in health, the two measures should be roughly equivalent. Large differences between CV and EV are possible, but they may also indicate deviations from the standard utility model, such as the loss aversion and

reference dependence models proposed by Tversky and Kahneman [26]. According to this alternative framework, individuals value changes with respect to a reference point (such as their status quo condition) and place a much higher (negative) weight on losses than they do on equivalent gains.

Data

As Smith and Pattanayak [7] note in their review of meta-analyses in nonmarket valuation, 'Synthesis requires the ability to define a common concept to be measured' (p. 274). The number of empirical health valuation studies has increased significantly over the last 10–15 years, but there is also significant heterogeneity in terms of the valuation approaches used and health effects evaluated across these studies. To define a meaningful common concept for meta-analysis, we used a similar approach to Johnson *et al.* [3].

In particular, we selected WTP values if

- they were estimated for well-defined acute health effects,
- they represented adults' private values for improving their own health,
- they were estimated using SP methods,
- the severity of the acute health effect could be expressed as and converted to a QWB score, and
- the change in duration of the acute effect could be quantified in terms of discrete days or episodes of illness.

We began by selecting the 53 observations included in the Johnson *et al.* study. These values were taken from five CV studies conducted in the United States in the late 1970s and 1980s. The studies were predominantly conducted for cardio-respiratory health effects associated with air pollution. We then conducted an extensive and detailed search and review of the health valuation literature to identify additional studies to include in the analysis. This search included other existing literature reviews [27–29] as well as several online databases (e.g. PubMed, ingenta, Econlit) using key words such as WTP, WTA, health value, health valuation, contingent valuation, and conjoint analysis. We also scanned the reference lists of these articles and consulted with other experts

in the field to identify additional candidate studies. Using these methods, we identified 183 additional value estimates (from 12 other studies), which expands the estimation sample by a factor of more than four. These additional studies were, for the most part, conducted after 1990, and they include research conducted both in the United States and in other countries. Appendix A provides more detailed descriptions of some of the key characteristics of these studies, including the valuation methods used, the health effects valued, the country studied, and the number of value estimates included in the meta-analysis.

As is commonly done in meta-regression analyses, we did not define overly restrictive quality criteria for including or excluding studies from the analysis. Rather, we followed the recommendation of Stanley [6] who writes 'when in doubt, it is best to err on the side of inclusion... Differences in quality, data, or methods do not provide a valid justification for omitting studies [from meta-analysis]' (p. 135). Excluding studies based on quality would require making judgments that might introduce more bias into the results than it would avoid. Moreover, to the extent possible, we have included explanatory variables in the analysis to control for methodological differences.

For the 236 selected values, Tables 2 and 3 describe and summarize the main variables that we used in the meta-analysis. WTPACUTE was the key variable of interest. It represents individuals' WTP to avoid or to reduce the number of days with a specific acute condition over the course of a year. All WTP estimates were converted to 2000 US dollars using the consumer price index (CPI), and, if they were originally measured in another currency, we first converted the estimates to dollars using the purchasing power parity (PPP) index. Most of the selected studies estimate and report average WTP values. If only median WTP values were reported, we included these estimates in WTPACUTE.

Following the approach used by Johnson *et al.*, we characterized the change in acute health outcomes associated with each WTP value in two main dimensions. First, we created the variable Δ DAYS to capture changes in the duration of the health effect. In most cases, this variable represents the reduction in the number of days over the course of a year that one experiences a given condition, such as shortness of breath, nausea, or headache. In a relatively small number of cases ($N = 17$), this variable represents a change in

Table 2. Descriptions of variables used in the meta-analysis

Variables	Description
WTPACUTE	Mean WTP for health change (in 2000 US dollars) ^a
ΔDAYS	Reduction in number of days with acute effect
ΔQWB	Improvement in health-related quality of life on affected days (= 1 – total QWB score)
QWBSYMSCORE	QWB symptom score (27 symptoms)
QWBMOBScore	QWB mobility score (3 levels)
QWBSACScore	QWB social activity score (3 levels)
QWBPACScore	QWB physical activity score (3 levels)
INCOME	Mean household income (in 2000 US dollars) ^a
AGE	Mean age
% MALE	Percent male
US	= 1 if study was conducted in the US; = 0 otherwise
WTPAVOID	= 1 if value was stated for avoiding a decrease in health; = 0 otherwise
OPEN ENDED	= 1 if an open-ended value elicitation method was used; = 0 otherwise
PAYMENT CARD	= 1 if a payment card value elicitation method was used; = 0 otherwise
IN PERSON	= 1 if survey was conducted as an in-person interview; = 0 otherwise
JOURNAL	= 1 if publication was published in a peer-reviewed journal; = 0 otherwise
SAMPLE SIZE	Number of respondents used to estimate the WTP value

^a Converted to dollars using the PPP index if in foreign currency; median WTP used if mean value not reported.

Table 3. Summary statistics for variables used in the meta-analysis

Variables	N	Mean	SD	Min	Median	Max
WTPACUTE	236	270.22	322.06	2.70	145.62	2927.69
ΔDAYS	236	11.9	20.5	1	5	90
ΔQWB	236	0.37	0.11	0.17	0.36	0.57
QWBSYMSCORE	235	0.26	0.03	0.17	0.257	0.30
QWBMOBScore	235	0.03	0.03	0	0	0.09
QWBSACScore	235	0.05	0.04	0	0.061	0.11
QWBPACScore	235	0.03	0.03	0	0	0.08
INCOME	236	46 348	13 618	21 891	47 067	88 020
AGE	236	45.2	6.8	35.4	44.5	68
% MALE	236	49.25	14.74	0	47.7	100
US	236	0.39	0.49	0	0	1
WTPAVOID	236	0.88	0.33	0	1	1
OPEN ENDED	236	0.14	0.35	0	0	1
PAYMENT CARD	236	0.29	0.46	0	0	1
IN PERSON	236	0.42	0.49	0	0	1
JOURNAL	236	0.67	0.47	0	1	1
SAMPLE SIZE	236	316.04	151.91	20	399	832

health that was expressed as a reduction in the number of acute events, such as asthma, angina, or allergy attacks. We assumed that each of these events was roughly equivalent to a day of acute illness, and in statistical tests of the regression results, the effect of ΔDAYS on WTP was not found to be significantly different for these observations.

Second, we used the QWB Scale to characterize the severity of the acute health effect. The QWB Scale characterizes health outcomes in four

dimensions – symptoms, mobility, social activity, and physical activity – each of which can be scored separately. Using the health effect descriptions from the valuation studies, we assigned each health effect to a defined level for all four dimensions. We then used the premeasured QWB weights [30] to assign a numerical score to each level.

The resulting four scores were captured in the variables QWBSYMScore, QWBMOBScore,